

Day 18 - Model / Method Comparison

Model Comparison Table

Model	Accuracy	Precision	Recall	F1-Score	BER
Logistic Regressi	0.742	0.667	0.681	0.674	0.258
SVM	0.758	0.846	0.468	0.603	0.242
Random Forest	0.708	0.63	0.617	0.624	0.292
MLP	0.75	0.717	0.596	0.651	0.25

Data Limitations

- Dataset size is relatively small (399 samples).
- Analog Value and Analog Voltage are highly correlated (redundant information).
- Limited number of features.
- Performance may vary under unseen NTU conditions.

Shortlisted Models for Deeper Experiments

Candidate 1: SVM

- Highest Accuracy
- Highest Precision
- Lowest BER
- Strong classification performance



Candidate 2 : Logistic Regression

- Highest Recall
- Highest F1-Score
- Simple and interpretable
- Good baseline model

